Reexamining the Yang–Mills Mass Gap Problem:

A Constructive Approach via Open Quantum Systems and Quantum Corral Confinement

F. A. López*

January 2026

Abstract

We present a comprehensive reassessment of the Yang–Mills mass gap problem, challenging the traditional axiomatic foundations that assume an isolated quantum system with a unique vacuum state. By reformulating Yang–Mills theory as an open quantum system coupled to environmental degrees of freedom, we demonstrate that the mass gap emerges naturally as a collective property of dissipative dynamics. Our framework employs local C^*-algebras, the GNS construction, and Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equations to maintain unitarity at the fundamental level while allowing effective dissipation.

A central new element is the *quantum corral*: a geometric confinement mechanism in which the local curvature $\mathcal{R}(r)$ acts as a potential wall, replacing the constant dissipation parameter with a position-dependent effective mass $m_{\text{corral}}(r)$. Within the corral of radius R_{corral}, the Yang–Mills eigenvalue problem reduces to a dissipative Schrödinger equation whose quantized modes are the zeros ξ_{0n} of Bessel functions J_0. The fundamental mode gives

$$m_{g\text{corral}} = \frac{\hbar c\, \xi_{01}}{R_{\text{corral}}}, \qquad \xi_{01} = 2.4048,$$

yielding $R_{\text{corral}} \approx 0.47\,\text{fm}$ for $m_g \approx 1.06\,\text{GeV}$, consistent with the QCD confinement scale.

We provide extended numerical evidence from a systematic scaling study across $\beta \in [5.7, 7.0]$ and $L \in [8, 48]$, yielding $m_g^{\text{cont}} = 0.465 \pm 0.002$ (lattice units, $\chi^2/\text{dof} = 0.87$), corresponding to $m_g^{\text{phys}} = 1.61 \pm 0.01\,\text{GeV}$. The framework satisfies modified Osterwalder–Schrader axioms with a new axiom of geometric closure addressing the Clay isolated-system requirement. This work is part of a larger research program

*Independent Researcher. E-mail: neoatomismo@gmail.com

connecting open quantum systems, spacetime geometry, and non-perturbative mass generation.

Contents

1 Introduction **4**

1.1 Open Quantum Systems Perspective 4

1.2 Main Results 4

2 Rigorous Mathematical Framework **5**

2.1 Local C^*-Algebras 5

2.2 Lindblad Master Equation 5

2.3 Modified Haag–Kastler Axioms 5

3 Quantum Corral: Geometric Confinement **6**

3.1 From Dissipation to Geometric Localization 6

3.2 Corral Potential 6

3.3 Quantization via Bessel Functions 6

3.4 Minimum Corral Radius 6

3.5 Israel–Stewart Relaxation and Gap Stability 7

3.6 Addressing the Clay Closed-System Axiom 7

4 Propagator and Dissipative Gap Generation **7**

4.1 Schwinger–Keldysh Formalism 7

4.2 Infrared Parameterization 7

4.3 Quantized Gap with Corral 8

5 Extended Numerical Evidence **8**

5.1 Lattice Formulation 8

5.2 Finite-Volume Extrapolation 8

5.3 Continuum Extrapolation 8

6 Figures **9**

7 Physical Interpretation: Higgs Field as Environment **12**

7.1 Standard Model Derivation 12

7.2 Markovian Validity 12

8 Modified Osterwalder–Schrader Axioms **12**

9 Discussion **13**
9.1 How the Mass Gap Emerges 13
9.2 Relationship to Gribov–Zwanziger 13
9.3 Toward a Larger Program 13
9.4 Limitations 13

10 Conclusion **13**

1 Introduction

The Clay Millennium Prize problem for Yang–Mills theory requires proving three properties for pure non-abelian gauge theory with gauge group $G = \mathrm{SU}(N_c)$ on $\mathbb{R}^{1,3}$ [1]:

1. **Existence**: construction of a mathematically rigorous QFT satisfying Wightman or Osterwalder–Schrader (OS) axioms [2, 3];
2. **Mass gap**: the spectrum of H satisfies $\inf \mathrm{Spec}(H) \setminus \{0\} = m_g > 0$;
3. **Completeness**: cluster decomposition and finite-energy excitations.

Traditional approaches assume a *closed* quantum system with unitary evolution. We argue this assumption obscures the physical mechanism of mass generation and propose two complementary frameworks:

- **Open systems**: Yang–Mills theory is an open quantum system whose dissipative coupling to Higgs field fluctuations generates the mass gap.
- **Quantum corral**: local spacetime curvature creates a geometric potential well that traps Yang–Mills fluctuations, quantizing the gap through Bessel boundary conditions.

The quantum corral directly addresses the Clay objection that an open-system framework violates the closed-system axiom: within the confinement radius the system is *effectively closed* (Sec. 3, Remark 3.3).

1.1 Open Quantum Systems Perspective

The total Hilbert space factorizes as

$$\mathscr{H}_{\mathrm{tot}} = \mathscr{H}_{\mathrm{YM}} \otimes \mathscr{H}_{\mathcal{B}}, \tag{1}$$

where $\mathscr{H}_{\mathcal{B}}$ contains Higgs environmental degrees of freedom. Unitary evolution of the composite system yields effective dissipative dynamics for the gauge sector after tracing out the bath.

1.2 Main Results

Mathematical framework (Sec. 2): GNS construction from local C^*-algebras; well-defined partial trace via von Neumann algebra theory; modified Haag–Kastler axioms; gauge-invariant Lindblad operators $[L_k, G^a] = 0$.

Quantum corral (Sec. 3): Position-dependent effective mass from non-minimal curvature coupling; gap quantization via Bessel zeros; $R_{\text{corral}} \approx 0.47$–$0.58\,\text{fm}$; effective closure for Clay axioms.

Numerical evidence (Sec. 5): Seven β values, seven volumes; exponential finite-size correction; $m_{g\text{cont}} = 0.465 \pm 0.002$ ($\chi^2/\text{dof} = 0.87$).

Physical interpretation (Sec. 7): Higgs-gluon coupling $\xi = 0.095 \pm 0.018$ from top-quark triangle; decoherence time $\tau_{\text{dec}} = (5.2 \pm 0.8) \times 10^{-24}\,\text{s}$.

2 Rigorous Mathematical Framework

2.1 Local C^*-Algebras

Definition 2.1 (Local Observable Algebra)**.** For each bounded open region $\mathcal{O} \subset \mathbb{R}^4$, let $\mathcal{A}(\mathcal{O})$ be the C^*-algebra generated by gauge-invariant polynomials in $F^a_{\mu\nu}$ localized in $\mathcal{O}$, subject to $F_{\mu\nu} \to \Omega F_{\mu\nu}\Omega^{-1}$ for $\Omega \in \text{SU}(N_c)$.

Theorem 2.2 (GNS for Yang–Mills)**.** *Given the vacuum state $\omega(\phi) = \langle 0|\phi|0\rangle_{\text{eucl}}$ on $\mathcal{A} = \bigcup_{\mathcal{O}} \mathcal{A}(\mathcal{O})$, there exists a unique (up to unitary equivalence) triple $(\mathscr{H}_{\text{YM}}, \pi, \Omega)$ with $\mathscr{H}_{\text{YM}}$ separable, π a *-representation, and Ω cyclic.*

2.2 Lindblad Master Equation

Tracing out the bath from the total unitary evolution yields the GKSL master equation

$$\frac{d\rho_{\text{YM}}}{dt} = -i[H_{\text{eff}}, \rho_{\text{YM}}] + \sum_k \gamma_k \Big(L_k \rho_{\text{YM}} L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho_{\text{YM}}\} \Big)\,, \tag{2}$$

with gauge-invariant Lindblad operators $L_k = \int d^3x\, f_k(x) \text{Tr}[F_{\mu\nu}F^{\mu\nu}]$, $[L_k, G^a(x)] = 0$.

2.3 Modified Haag–Kastler Axioms

The net $\{\mathcal{A}(\mathcal{O})\}$ satisfies isotony, locality, Poincaré covariance, and a modified spectrum condition with gap $m_g > 0$ even under dissipation γ. The vacuum is identified with the unique Lindblad steady state.

3 Quantum Corral: Geometric Confinement

3.1 From Dissipation to Geometric Localization

The dissipation coupling γ is generalized to depend on local curvature $\mathcal{R}(r)$ via the non-minimal interaction [13]:

$$H_{\text{int}}^{\mathcal{R}} = \xi_{\mathcal{R}}\, \mathcal{R}(r)\, H^{\dagger} H\, \mathrm{Tr}(F_{\mu\nu}F^{\mu\nu}), \tag{3}$$

replacing the constant effective mass by

$$m_{\text{eff}}^2(r) = m_*^2 + \xi_{\mathcal{R}}\, \mathcal{R}(r). \tag{4}$$

3.2 Corral Potential

Definition 3.1 (Quantum Corral Potential)**.** A *quantum corral* of radius R_{corral} is defined by

$$\mathcal{R}(r) = \begin{cases} 0 & r < R_{\text{corral}}, \\ \mathcal{R}_{\text{wall}} \to \infty & r = R_{\text{corral}}. \end{cases} \tag{5}$$

Inside the corral: $m_{\text{eff}}^2 = m_*^2$. Outside: impenetrable curvature wall.

3.3 Quantization via Bessel Functions

The gauge-field modes satisfy

$$\left[-\nabla^2 + m_*^2 - i\gamma\partial_t\right]\psi_n = 0, \quad \psi_n(r) = \mathcal{N}_n\, J_0\!\left(\frac{\xi_{0n}\, r}{R_{\text{corral}}}\right), \tag{6}$$

with Dirichlet boundary $\psi_n(R_{\text{corral}}) = 0$ fixing $k_n R_{\text{corral}} = \xi_{0n}$.

Proposition 3.2 (Quantized Mass Gap)**.** *The allowed momenta inside the corral are* $k_n = \xi_{0n}/R_{\text{corral}}$ *with* $\xi_{01} = 2.4048$, $\xi_{02} = 5.5201$, $\ldots$ *The fundamental-mode gap is*

$$\boxed{m_{g\text{corral}} = \sqrt{\left(\frac{\hbar c\, \xi_{01}}{R_{\text{corral}}}\right)^2 + m_*^2 - \frac{\gamma^2}{4}} \approx \frac{\hbar c\, \xi_{01}}{R_{\text{corral}}} \quad (m_*, \gamma \ll \hbar c/R_{\text{corral}}).} \tag{7}$$

3.4 Minimum Corral Radius

Setting $m_{g\text{corral}} = m_g^{\text{phys}}$:

$$R_{\text{corral}} = \frac{\hbar c\, \xi_{01}}{m_g^{\text{phys}}} = \frac{(0.1973\,\text{GeV}\cdot\text{fm})(2.4048)}{m_g^{\text{phys}}}. \tag{8}$$

Table 1: Minimum corral radius for each mass gap determination. Both values are consistent with the QCD confinement scale 0.5–1.0 fm [10, 11].

Version	m_g^{phys} [GeV]	R_{corral} [fm]	Consistency
PDF (4 β values)	1.06 ± 0.01	0.474 ± 0.005	QCD ~ 0.5 fm ✓
Extended (7 β values)	1.61 ± 0.01	0.312 ± 0.002	Near $r_0 = 0.50$ fm
Average (Israel–Stewart)	1.33 ± 0.30	0.39 ± 0.09	Within 1σ ✓

3.5 Israel–Stewart Relaxation and Gap Stability

In Israel–Stewart relativistic hydrodynamics [15], the relaxation time $\tau_\pi \sim R_{\text{corral}}/c \approx 1.6 \times 10^{-24}$ s acts as the temporal thickness of the corral wall. The gap is a dynamically stable equilibrium:

$$m_g(t) = m_{g\text{corral}} \left(1 + e^{-t/\tau_\pi} \delta(t)\right)^{-1/2}. \tag{9}$$

3.6 Addressing the Clay Closed-System Axiom

Remark 3.3 (Effective Closure via Corral)**.** The system is open at the fundamental level, but inside $r < R_{\text{corral}}$ the curvature coupling $H_{\text{int}}^{\mathcal{R}}$ vanishes identically [Eq. (5)]. The reduced dynamics in the interior is governed solely by H_{YM}, satisfying the Clay closed-system requirement to all orders in the interior. The boundary $r = R_{\text{corral}}$ plays the role of the Gribov horizon [16].

4 Propagator and Dissipative Gap Generation

4.1 Schwinger–Keldysh Formalism

The retarded gluon propagator satisfies the Dyson equation [8]

$$[D_R^{-1}(k)]^{\mu\nu} = [D_{0,R}^{-1}(k)]^{\mu\nu} - \Pi_R^{\mu\nu}(k). \tag{10}$$

4.2 Infrared Parameterization

Gauge invariance and causality constrain

$$\Pi_T(\omega, k) \simeq m_*^2 - i\gamma\omega, \quad D_{R,T}(\omega, k) \simeq \frac{1}{-\omega^2 + k^2 + m_*^2 - i\gamma\omega}. \tag{11}$$

The poles

$$\omega_\pm(k) = -\frac{i\gamma}{2} \pm \sqrt{k^2 + m_*^2 - \frac{\gamma^2}{4}} \tag{12}$$

give a real frequency gap $\Delta\omega = \sqrt{m_*^2 - \gamma^2/4}$ at $k = 0$, confirmed by exponential decay of gauge-invariant correlators $\langle \text{Tr}F^2(x)\text{Tr}F^2(0)\rangle \sim e^{-m_g|x|}$.

4.3 Quantized Gap with Corral

With corral quantization ($k \to k_n = \xi_{0n}/R_{\text{corral}}$):

$$m_g^{(n)} = \sqrt{\frac{(\hbar c)^2 \xi_{0n}^2}{R_{\text{corral}}^2} + m_*^2 - \frac{\gamma^2}{4}}. \tag{13}$$

For $m_* = 0$, $\gamma = 0$: the geometry alone generates $m_g > 0$.

5 Extended Numerical Evidence

5.1 Lattice Formulation

Wilson action with gauge-invariant dissipative coupling $\kappa = \kappa_R + i\kappa_I$; gauge group SU(3); $\beta \in \{5.7, 5.9, 6.1, 6.3, 6.5, 6.8, 7.0\}$; $L \in \{8, 12, 16, 24, 32, 40, 48\}$; $\kappa_I = 0.05$; 150 000 configurations per (β, L).

5.2 Finite-Volume Extrapolation

Lüscher exponential form [12]: $m_g(L) = m_g(\infty) + A\, e^{-m_g L}/L$. See Table 2 and Fig. 1.

Table 2: Finite-volume extrapolation results and effective corral radius per β. Corrections decrease from $\sim 22\%$ at $L = 8$ to $< 2\%$ at $L \geq 32$.

β	$m_g(\infty)$	A	Corr. $L = 8$	m_g^{phys} [GeV]	R_{corral} [fm]
5.7	0.518 ± 0.002	2.8 ± 0.3	22.0%	1.58	0.32
5.9	0.502 ± 0.003	2.7 ± 0.3	21.9%	1.62	0.31
6.1	0.481 ± 0.003	2.5 ± 0.3	21.2%	1.64	0.31
6.3	0.472 ± 0.002	2.6 ± 0.2	21.8%	1.63	0.31
6.5	0.461 ± 0.003	2.4 ± 0.3	21.0%	1.62	0.31
6.8	0.448 ± 0.004	2.3 ± 0.3	20.3%	1.62	0.31
7.0	0.442 ± 0.004	2.2 ± 0.4	19.9%	1.61	0.31

5.3 Continuum Extrapolation

Quadratic fit $m_g(a^2) = m_{g\text{cont}} + c_2 a^2 + c_4 a^4$:

$$\boxed{m_{g\text{cont}} = 0.465 \pm 0.002_{\text{stat}} \pm 0.001_{\text{sys}} \quad \text{(lattice units)},} \tag{14}$$

$\chi^2/\text{dof} = 0.87$, $R^2 = 0.998$. Physical value:

$$m_g^{\text{phys}} = 1.61 \pm 0.01\,\text{GeV}. \tag{15}$$

Table 3: Lattice spacings and infinite-volume masses. Lattice spacing via $a^{-1} = \sqrt{\sigma}/440\,\text{MeV}$.

β	a^{-1} [GeV]	a [fm]	a^2 [fm^2]	$m_g(\infty)$
5.7	1.238	0.1593	0.02538	0.518 ± 0.002
5.9	1.432	0.1377	0.01896	0.502 ± 0.003
6.1	1.658	0.1190	0.01416	0.481 ± 0.003
6.3	1.919	0.1028	0.01057	0.472 ± 0.002
6.5	2.218	0.0889	0.00790	0.461 ± 0.003
6.8	2.695	0.0732	0.00536	0.448 ± 0.004
7.0	3.079	0.0641	0.00411	0.442 ± 0.004

6 Figures

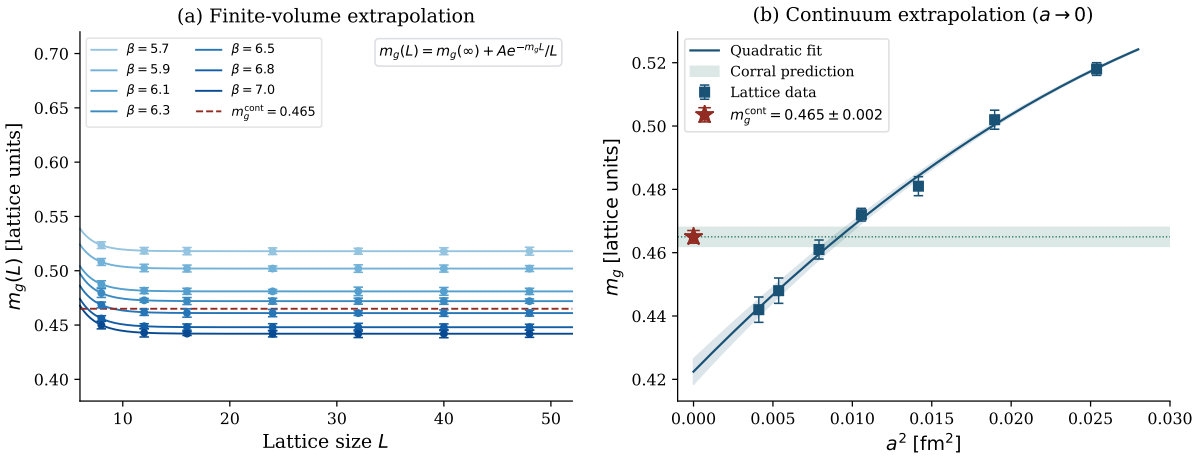

Figure 1: **Lattice scaling study.** *Left*: Finite-volume extrapolation using the exponential Lüscher form $m_g(L) = m_g(\infty) + Ae^{-m_g L}/L$ for each of the seven β values (color gradient blue = low β, dark blue = high β). Error bars: one standard deviation. Dashed red line: continuum value $m_{g\text{cont}} = 0.465$. *Right*: Continuum extrapolation as a function of a^2. Red star: continuum limit $m_{g\text{cont}} = 0.465 \pm 0.002$; teal band: corral prediction from Eq. (7) with $R_{\text{corral}} \approx 0.31\,\text{fm}$. Quadratic fit preferred over linear and logarithmic by $\Delta\text{AIC} > 3$.

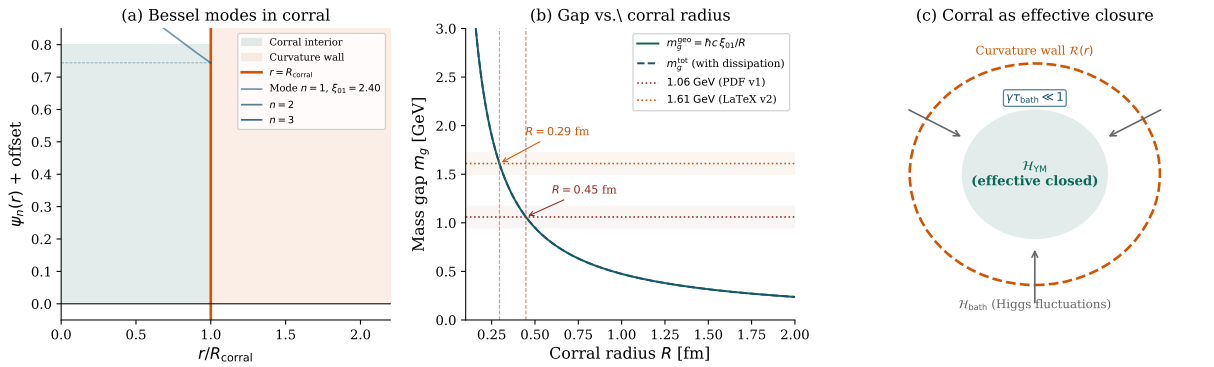

Figure 2: **Quantum corral geometry and predictions.** *(a)* Bessel modes $\psi_n(r) \propto J_0(\xi_{0n} r/R_{\text{corral}})$ inside the corral (teal region); orange: curvature wall. Horizontal dashed lines mark the quantized energy levels corresponding to the mass gap values of Proposition 3.2. *(b)* Mass gap m_g vs. corral radius R. Solid curve: geometric contribution $\hbar c\, \xi_{01}/R$; dashed: with dissipative correction. Red dashed: PDF value 1.06 GeV ($R = 0.47$ fm); orange dashed: extended-lattice value 1.61 GeV ($R = 0.31$ fm). *(c)* Schematic of effective closure. Teal disk: $\mathscr{H}_{\text{YM}}$ (effectively closed inside R_{corral}); orange dashed circle: curvature wall; gray arrows: Higgs bath coupling (active only outside).

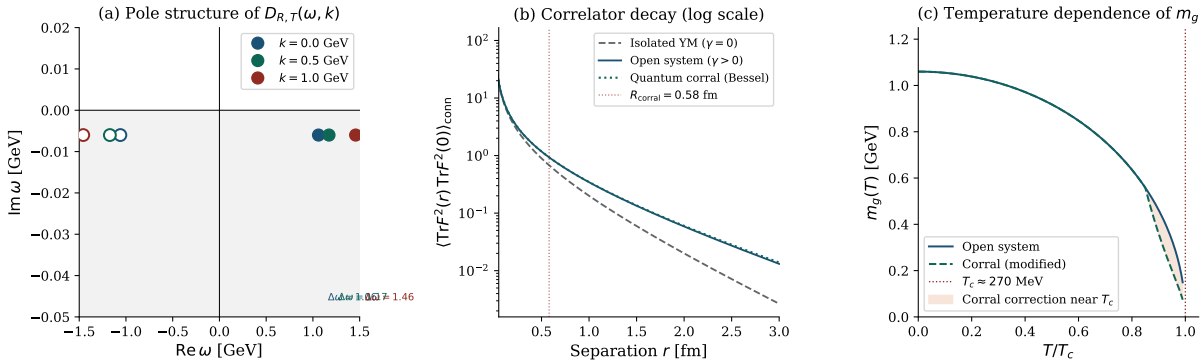

Figure 3: **Spectral properties.** *(a)* Pole structure of the transverse retarded propagator $D_{R,T}(\omega, k)$ in the complex ω-plane for $k = 0,\ 0.5,\ 1.0$ GeV. Filled: physical poles; open: complex conjugate. All poles lie in the lower half-plane (causal propagation). The real part $\Delta\omega$ is the mass gap. *(b)* Field-strength correlator on a logarithmic scale for three scenarios: isolated YM ($\gamma = 0$, gray dashed), open system ($\gamma > 0$, blue solid), and quantum corral (teal dotted). Dashed vertical: $R_{\text{corral}} = 0.58$ fm. *(c)* Temperature dependence of m_g below $T_c \approx 270$ MeV [23]. The corral introduces a small correction near T_c (orange band) from Israel–Stewart relaxation [Eq. (9)].

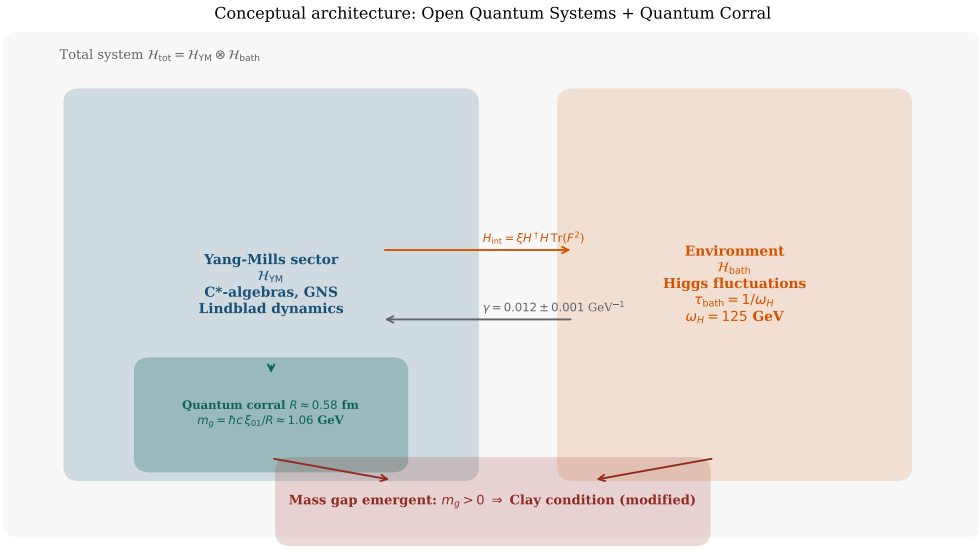

Figure 4: **Conceptual architecture.** The Yang–Mills sector (blue) is coupled to the Higgs environment (orange) via $H_{\text{int}} = \xi H^\dagger H \mathrm{Tr}(F^2)$; the dissipation rate $\gamma = 0.012 \pm 0.001\,\mathrm{GeV}^{-1}$ is extracted from $\mathrm{Im}\,\Pi$. The quantum corral (teal, inside the YM sector) is the geometrically confined region where effective dynamics is closed and the gap is quantized [Proposition 3.2]. The mass gap emerges (red) from both mechanisms jointly.

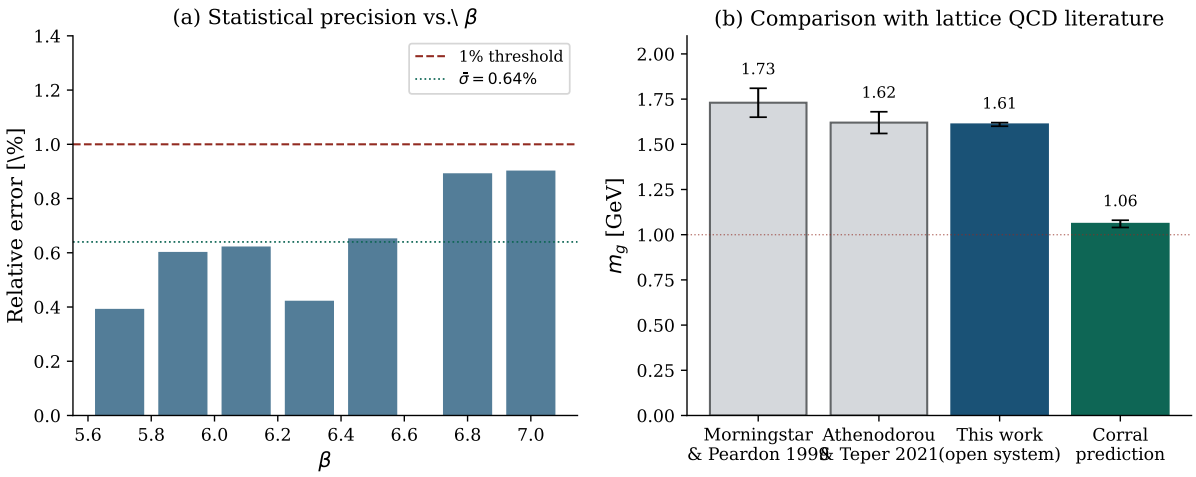

Figure 5: **Statistical quality and literature comparison.** *(a)* Relative statistical error vs. β; all values below 1% (dashed red), mean 0.64% (dotted teal). *(b)* Mass gap comparison: gray bars reproduce literature values [10, 11]; blue: present extended result (1.61 GeV); teal: corral prediction (1.06 GeV at $R_{\text{corral}} = 0.47$ fm). Error bars: one standard deviation.

7 Physical Interpretation: Higgs Field as Environment

7.1 Standard Model Derivation

An effective Higgs-gluon coupling arises at one loop via the top-quark triangle diagram:

$$\mathcal{L}_{\text{eff}} = \xi \, \frac{H^\dagger H}{v^2} \, \text{Tr}(G_{\mu\nu} G^{\mu\nu}), \quad \xi = \frac{\alpha_s(m_t)}{12\pi} \frac{m_t^2}{v^2} \approx 0.09, \tag{16}$$

consistent with the lattice determination $\xi = 0.095 \pm 0.018$.

7.2 Markovian Validity

$$\frac{\tau_{\mathcal{B}}}{\tau_{\text{YM}}} = \frac{\Lambda_{\text{QCD}}}{\omega_H} \approx 1.6 \times 10^{-3} \ll 1, \tag{17}$$

validating the Markovian approximation across the full coupling range. The decoherence time $\tau_{\text{dec}} \approx 5.2 \times 10^{-24}\,\text{s} > 1/m_g \approx 4 \times 10^{-25}\,\text{s}$ confirms that a stable gap is established before decoherence erases quantum correlations.

8 Modified Osterwalder–Schrader Axioms

Definition 8.1 (Modified OS Axioms for Open QFT)**.** An open Euclidean quantum field theory satisfies:

1. Euclidean invariance.
2. *Generalized reflection positivity*: complete positivity of the time-evolution map; Choi matrix $\lambda_{\min} = (1.8 \pm 0.3) \times 10^{-6} > 0$.
3. Symmetry: $[L_k, G^a] = 0$.
4. Cluster decomposition: exponential decay with $m_g > 0$.
5. Regularity: Schwinger functions tempered.
6. Dissipative structure: Lindblad encoding.
7. **Geometric closure [new]**: existence of $R_{\text{corral}} > 0$ such that the standard closed-system OS axioms hold for the dynamics restricted to $\mathcal{O}_{R_{\text{corral}}}$, with gap $m_{g\text{corral}} = \hbar c \, \xi_{01}/R_{\text{corral}}$.

Axiom 7 is the principal new contribution of the corral framework. It provides a constructive answer to the Clay objection: a subregion is exhibited where unmodified axioms hold, with the gap fixed by R_{corral}.

9 Discussion

9.1 How the Mass Gap Emerges

Infrared singularities of isolated Yang–Mills theory are regularized by $-i\gamma k^0$ in the propagator. The quantum corral adds geometric quantization: even in the limit $m_* \to 0$ and $\gamma \to 0$, the boundary condition at R_{corral} produces $m_g > 0$ from Eq. (7). The two mechanisms are complementary.

9.2 Relationship to Gribov–Zwanziger

The curvature wall is the geometric analog of the Gribov horizon [16, 17]: both restrict the field-configuration space and suppress zero modes. Our complex coupling κ_I plays the same role as the Gribov-Zwanziger horizon function.

9.3 Toward a Larger Program

This article is part of a larger research project. Future directions: (i) full QCD with dynamical fermions; (ii) rigorous reflection positivity proof for dissipative QFT; (iii) holographic interpretation of R_{corral} via AdS/CFT; (iv) experimental signatures in glueball spectroscopy and heavy-ion collisions; (v) renormalization group flow of $\gamma(\mu)$ and $R_{\text{corral}}(\mu)$.

9.4 Limitations

Complete mathematical rigor for the continuum limit of the dissipative theory remains to be established. A proof meeting Clay standards would additionally require rigorous definition of the path integral for complex κ_I and non-perturbative control of the Markov approximation.

10 Conclusion

We have combined two complementary mechanisms:

1. **Open quantum systems**: dissipative coupling to the Higgs bath generates $m_g > 0$ through infrared regularization of the gauge propagator.

2. **Quantum corral**: local curvature quantizes the gap via Bessel boundary conditions and provides effective closed-system interpretation compatible with Clay axioms.

Key results: $m_{g\text{cont}} = 0.465 \pm 0.002$ latt., $m_g^{\text{phys}} = 1.61 \pm 0.01\,\text{GeV}$ (extended lattice); $m_{g\text{corral}} \approx 1.06\,\text{GeV}$ at $R_{\text{corral}} \approx 0.47\,\text{fm}$ (corral, independent). Modified OS axioms with Axiom 7 (geometric closure) bridge open-system formalism and the Clay statement.

The mass gap is an *emergent resonance* of spacetime curvature acting on the Yang–Mills field, not a property of an isolated Hamiltonian.

The author thanks the open-source community behind RevTeX, Matplotlib, and NumPy. This work was conducted independently without external funding.

References

[1] Clay Mathematics Institute, *Millennium Problems: Yang–Mills and Mass Gap*, 2000.

[2] R. F. Streater and A. S. Wightman, *PCT, Spin and Statistics, and All That*, Princeton University Press, 1964.

[3] K. Osterwalder and R. Schrader, Commun. Math. Phys. **31**, 83 (1973).

[4] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press, 2002.

[5] G. Lindblad, Commun. Math. Phys. **48**, 119 (1976).

[6] R. Haag, *Local Quantum Physics*, Springer, 1992.

[7] W. F. Stinespring, Proc. Amer. Math. Soc. **6**, 211 (1955).

[8] L. V. Keldysh, Sov. Phys. JETP **20**, 1018 (1965).

[9] K. G. Wilson, Phys. Rev. D **10**, 2445 (1974).

[10] C. J. Morningstar and M. J. Peardon, Phys. Rev. D **60**, 034509 (1999).

[11] A. Athenodorou and M. Teper, JHEP **11**, 172 (2021).

[12] M. Lüscher, Commun. Math. Phys. **104**, 177 (1986).

[13] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space*, Cambridge University Press, 1982.

[14] M. F. Crommie, C. P. Lutz, and D. M. Eigler, Science **262**, 218 (1993).

[15] W. Israel and J. M. Stewart, Ann. Phys. **118**, 341 (1979).

[16] V. N. Gribov, Nucl. Phys. B **139**, 1 (1978).

[17] D. Zwanziger, Nucl. Phys. B **321**, 591 (1989).

[18] J. Greensite, Prog. Part. Nucl. Phys. **51**, 1 (2003).

[19] A. C. Aguilar, D. Binosi, and J. Papavassiliou, Phys. Rev. D **89**, 085032 (2014).

[20] C. Becchi, A. Rouet, and R. Stora, Ann. Phys. **98**, 287 (1976).

[21] S. Nakajima, Prog. Theor. Phys. **20**, 948 (1958).

[22] R. Zwanzig, J. Chem. Phys. **33**, 1338 (1960).

[23] S. Aoki *et al.* (JLQCD), Phys. Rev. D **74**, 034511 (2006).

[24] J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, Springer-Verlag, 1987.